

Sri Lanka Institute of Information Technology

Enterprise Publishing Management System

Business Architecture Assessment Report

IE3121 – Enterprise Architecture

Project ID: ISE_WE_0101_22

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1. Executive Summary

The Book Publishing House is an enterprise whose responsibility for managing the complete publishing process. Including author registration, manuscript submission, editorial review, proofreading, book production, printing, inventory management, customer sales, and royalty payments. Currently, many of these business processes rely on manual processes such as paper-based records, spreadsheets, emails, and phone calls, resulting in inefficient workflows, limited collaboration, delayed decision-making, and fragmented information across departments.

This Business Architecture Assessment analyzes the current business environment to identify operational challenges and determine how Enterprise Architecture can support its digital transformation objectives. The assessment focuses on analyzing and reviewing the existing business processes, stakeholder requirements, organizational capabilities, and business operations to identify areas that requiring improvement. It also analyzes how an integrated enterprise system can improve the organizational efficiency and support long-term business growth.

They are several key issues, including the absence of centralized information management, limited visibility into manuscript progress, inefficient editorial coordination, manual inventory management, fragmented customer order processing, manual royalty calculations, and a lack of real-time reporting for management. All these problems have negative implications for the productivity of operations, communications, service delivery, and management decisions.

To resolve the above-mentioned problems, the proposed Enterprise Publishing Management System (EPMS) provides a centralized, web-based platform that integrates the organization's core business functions into a single enterprise solution. The proposed future business architecture introduces standardized workflows, secure role-based access control, automated business processes, centralized data management, and real-time reporting capabilities. This digital transformation is expected to improve operational efficiency, strengthen collaboration among stakeholders, enhance information security, increase customer and author satisfaction, and provide a scalable foundation for future organizational growth.

2. Organization Profile

Reading Planet Publishers is a small book publisher in the book publishing industry. Services offered include book publishing, distribution, and book selling. It publishes books by authoring, reviewing manuscripts, proofreading, producing books, inventory management, and selling books to customers.

It has a simple organizational structure with a Managing Director/Owner and various departments including Editorial, Production and Design, Marketing and Sales, Distribution, Finance and Administration. These departments are interdependent in their role of publishing and selling books.

At present Reading Planet Publishers works largely on a manual basis. Paper records and spreadsheets are used to keep track of the records and communications are primarily through email and phone calls. There is no website or integrated information system presently in the organization. This is why the challenges are inefficient information management, manuscript and inventory tracking, delays with customer order management and royalty payments.

The organization wants to enhance these processes by implementing a web-based Enterprise Publishing Management System (EPMS). The proposed system will provide centralized access to important information, automate selected business processes, enhance communication and will give better access to information to management, employees, authors and customers.

3. Business Problem Statement

The primary issue with Reading Planet Publishers is that they don't have a central repository for information. Paper records and spreadsheets are maintained in each department, with duplicate and often conflicting information. Time is being wasted by staff in checking and correcting records and management is unable to obtain one true picture of the business.

This will have an impact on the editorial process directly. After submission of the manuscript, there is no good way to follow it. Authors have to call or email to inquire about their submission and editors have to go through email threads and printed drafts to locate the latest version. Outdated copies can lead to reviewers and proofreaders having to repeat effort to revise articles and delay publication.

The inventory and sales are done manually. Stock is maintained by hand and is often wrong, so orders are taken for books that are out of stock and reprints come too late. All sales are by telephone or face to face, creating a time constraint for the customer and restricting sales to local customers and orders are processed manually by hand thus increasing the likelihood of error.

They are basing the royalty payments on sales figures and doing so manually at the end of each accounting period. This is a slow and error-prone process and mistakes in author payments result in conflicts and loss of trust.

Management is also deficient in information to aid decision making. Data is not consolidated and is manually updated, it is not possible to generate reports on sales, stock and title performance prior to the transaction and the data is out of date by the time it is available. Furthermore, no control over who has access to information, manuscripts, contracts and financial records are shared with all staff and there is no record of changes.

These issues cause loss in efficiency, delay publication and restrict growth. They support the proposed Enterprise Publishing Management System (EPMS), a Web-based centralized platform that will integrate key business functions of the organization with automated workflows, role-based access control, and real time reporting.

4. Stakeholder Analysis

4.1 Stakeholder Identification

Table 4.1: Stakeholder Identification

Stakeholder	**Internal / External**	**Role**
Authors	External	Content Creators
Editorial Team (Editors & Proofreaders)	Internal	Content Quality Managers
Production Team (Designers & Production Managers)	Internal	Book Producers
Operations Team (Inventory & Sales Staff)	Internal	Logistics Operations
Finance Staff	Internal	Financial Administrators
Administrator / Management	Internal	Business & System Leaders
Customers / Bookstores	External	End Users & Retail Outlets

4.2 Stakeholder Matrix

Table 4.2: Stakeholder Matrix

Stakeholder	**Role**	**Interest**	**Influence**	**Impact**	**Engagement Strategy**
Authors	External Content Creators	Submit manuscripts, track status, approve cover design, receive royalty payments	Medium	High	Automated email notifications, user-friendly portal dashboard, clear

					contract terms
Editorial Team (Editors & Proofreaders)	Internal Content Quality Managers	Review manuscripts, request author revisions, perform proofreading & formatting	High	High	Centralized editorial workflow dashboard, version history tools
Production Team (Designers & Production Managers)	Internal Book Producers	Handle layout, cover design, send for printing, manage printing schedules	Medium	High	Milestone tracking boards, shared file repositories for designs
Operations Team (Inventory & Sales Staff)	Internal Logistics Operations	Track warehouse stock, manage pre-orders/purchases, fulfill customer deliveries	Low	High	Real-time inventory status, automated re-order alerts, integrated storefront
Finance Staff	Financial Administrators	Calculate author royalties, process payments, generate sales reports	Medium	High	Automated royalty engines, clear sales transaction logs

Administrator / Management	Business & System Leaders	Overall operational efficiency, system security (RBAC), business analytics	High	High	Executive summary reports, role-based access management, security auditing
Customers / Bookstores	End Users & Retail Outlets	Place pre-orders and purchases, track book delivery status	Low	Medium	Online payment gateway, delivery updates

4.3 Stakeholder Classification

The stakeholders of the Enterprise Publishing Management System are classified based on their level of interest and influence in the project. The Administrator/Management and the Editorial Team have high interest and high influence and therefore require close management and continuous engagement. Authors, the Production Team, and Finance Staff have moderate levels of interest or influence and should be kept satisfied through appropriate communication and system support. The Operations Team and Customers/Bookstores have lower levels of influence and can be monitored while ensuring that their operational and service requirements are adequately addressed.

Power–Interest Grid

Table 4.3: Power–Interest Grid

	High Interest	Low Interest
High Power / Influence	Manage Closely • Administrator/Management • Editorial Team	Keep Satisfied • Finance Staff

Low Power / Influence	Keep Informed • Authors • Production Team • Operations Team	Monitor • Customers/Bookstores

4.4 Stakeholder Impact Analysis

The implementation of the Enterprise Publishing Management System is expected to have a significant impact on both internal and external stakeholders. Administrator/Management and the Editorial Team are the most influential stakeholders because they have high decision-making authority and strong involvement in the project. Management will be responsible for overseeing the overall system of adoption, while the Editorial Team will have a major influence on the manuscript review and publishing workflow.

Authors are among the most affected stakeholders because the system will replace several manual activities with digital processes. They will use the system for manuscript submission, submission tracking, revision handling, contract viewing, and accessing royalty information. The Editorial and Production Teams will also experience significant operational changes through centralized workflow management, document version control, design coordination, and production tracking. Operations and Finance Staff will be affected through improved inventory, sales, royalty, and financial processes. Customers and Bookstores will experience a moderate impact through improved purchasing, pre-ordering, and delivery-related services.

Some resistance to change may occur, particularly among employees who are accustomed to existing paper-based, email, telephone, and spreadsheet-based processes. Users may initially require training and support to adapt to the new system. To address this resistance, the project should provide clear communication, user guidance, demonstrations, and appropriate training before system adoption.

Effective communication with stakeholders is essential throughout the project. Management should receive regular progress updates and decision-related information, while operational teams should be consulted regarding process requirements and system changes. Authors and external users should receive clear instructions, system guidance, and relevant updates regarding their activities. This stakeholder engagement will support smoother adoption and help ensure that the EPMS achieves its intended organizational benefits.

5. **Define the process scope**

EPIC 1 – Author & Manuscript Management

Table 5.1: Author & Manuscript Management

Item	**Student response**
Process name	Author & Manuscript Management Process
Triggering event	An author prepares a manuscript and initiates the submission process using the existing manual submission methods.
Start point	The author prepares the manuscript and begins the submission process.
End point or points	1.The manuscript is submitted to the publishing house and the relevant manuscript information is handed over to the Editorial Team for editorial processing. 2. When a revision is requested, the author prepares and resubmits a revised manuscript version for further editorial processing.
Main business objective	To manage the author-side manuscript submission lifecycle by enabling authors to prepare and submit manuscripts and supporting files, track submission progress, respond to revision requests, resubmit revised manuscript versions, and maintain access to manuscript-related information until the manuscript is handed over to the editorial process.

Items within scope	• Prepare and submit manuscript • Enter manuscript information • Provide manuscript and supporting documents • Track manuscript submission status • View manuscript details and submission history • Receive revision requests • Prepare revised manuscript • Resubmit revised manuscript • Maintain manuscript version history • View contract information • View royalty information • Handover submitted manuscript information to the Editorial Team
Items outside scope	• Editorial review and manuscript processing • Editorial decisions • Proofreading and editing • Reviewer/editor assignment • Book design and page layout • Printing operations • Inventory management • Customer orders and sales • Book delivery and distribution • Contract management • Royalty calculation and payment processing

EPIC 2 – Editorial Workflow & Book Production

Table 5.2: Editorial Workflow & Book Production

Item	**Student response**
Process name	Editorial Workflow & Book Production Process
Triggering event	An accepted manuscript is received from Epic 1 – Author & Manuscript Management for editorial processing.
Start point	The Chief Editor receives an accepted manuscript from Epic 1 and assigns it to an editor for review.
End point or points	1. The manuscript is rejected following editorial review, ending the editorial workflow. 2. The print-ready book is approved, marked ready for printing, and transferred to the Printing Department (Epic 3).
Main business objective	To manage the editorial review and book production process by reviewing accepted manuscripts, coordinating revisions, preparing book designs and layouts, performing quality checks, obtaining author approval, granting production approval, and producing a print-ready book for the next stage of publishing.
Items within scope	• Receive accepted manuscript from Epic 1 • Assign editor • Review manuscript • Send revision request (when required) • Receive revised manuscript • Approve editorial decision • Reject manuscript

	• Create cover design • Prepare page layout • Revise design/layout • Perform production quality check • Review book design • Grant production approval • Mark book ready for printing • Transfer print-ready book to the Printing Department (Epic 3)
Items outside scope	• Author registration • Manuscript submission • Manuscript acceptance (Epic 1) • Printing operations • Book publishing and distribution • Inventory management • Customer orders and sales • Book delivery and distribution • Royalty calculation and finance management

EPIC 3 – Publishing Operations (Printing, Inventory, Sales & Distribution)

Table 5.3: Publishing Operations

Item	**Student response**
Process name	Publishing Operations (Printing, Inventory, Sales & Distribution) process
Triggering event	A book is approved for printing and is marked as Ready for Printing.

Start point	The process starts when the Production Department receives the approved book and printing details.
End point or points	The process ends when printed books are added to inventory, customer and wholesale orders are delivered, and operational sales data is available for the next process.
Main business objective	To manage the operational lifecycle of published books from printing through inventory, sales, order fulfillment, delivery and wholesale distribution.
Items within scope	• Creating and managing print jobs • Printing books • Checking printed books • Recording printed stock • Managing inventory Browse published books • Search books • Filter books • View book details • Add books to cart • Update cart quantities • Remove cart items • View order history • Track orders • Confirm customer orders • Reserve inventory • Cancel eligible orders • Receiving customer orders • Checking stock availability • Processing wholesale orders • Approving wholesale orders

	• Packing orders • Dispatching orders • Delivering orders • Handling incorrect or damaged deliveries • Replenishing stock when required
Items outside scope	• Author registration • Manuscript submission • Editorial review • Proofreading • Royalty calculation • Author payments • User registration and authentication • Finance processes unrelated to publishing operations

EPIC 4 – Administration, Finance & Royalty Management

Table 5.2: Administration, Finance & Royalty Management

Item	**Student response**
Process name	Administration, Finance & Royalty Management
Triggering event	The end of a royalty period is reached , requiring royalties to be calculated for all authors with active agreements
Start point	Finance Officer requests the period's sales records from the Sales & Distribution Department
End point or points	1) Royalty paid, statement issued to author, and payment filed in the ledger. 2) Payment deferred to the next period due to insufficient funds. 3) Payment rejected by the Managing Director and returned to Finance
Main business objective	To calculate author royalties accurately from recorded sales, obtain the required approvals, pay authors on time, and produce financial reports for management

Items within scope	• Collecting sales figures • retrieving royalty agreements • verifying royalty rates • calculating royalty amounts • preparing royalty statements • internal review • executive approval of high value payments • fund availability check • payment issue • author confirmation and dispute handling • recording and filing • financial report preparation
Items outside scope	• Customer order taking and sales transactions • manuscript submission and editorial review • book production, printing and inventory • author registration and contract negotiation • staff payroll and tax filing • login, user accounts and access control

6. Identify the swim lanes

EPIC 1 – Author & Manuscript Management

Table 6.1: Administration, Finance & Royalty Management

Swimlane	**Main responsibilities in the process**
Author	Prepare and submit manuscripts, enter manuscript information, provide manuscript and supporting documents, track submission status, receive revision requests, prepare revised manuscript versions, resubmit revised manuscripts, and view manuscript-related information.

Editorial Team	Receive submitted manuscripts, review manuscripts, request revisions when required, receive revised manuscripts, and communicate the editorial outcome to the Author.

EPIC 2 – Editorial Workflow & Book Production

Table 6.2: Editorial Workflow & Book Production

Swimlane	**Main responsibilities in the process**
Chief Editor	Receive accepted manuscripts, assign editors, receive revised manuscripts, and approve editorial decisions.
Editor	Review manuscripts, evaluate review outcomes, approve or reject manuscripts, and send revision requests when revisions are required.
Designer	Create book cover designs, prepare page layouts, and revise designs or layouts based on quality check results or requested changes
Production Manager	Perform production quality checks, grant production approval, mark books ready for printing, and transfer print-ready books to the Printing Department
Author	Review the completed book design and either approve it or request design changes.

EPIC 3 – Publishing Operations (Printing, Inventory, Sales & Distribution)

Table 6.3: Publishing Operations

Swimlane	**Main responsibilities in the process**
Production Staff	Create and schedule print jobs, assign printers, update printing status, record printed quantities and complete print jobs.
Warehouse Staff	Receive printed stock, update and adjust inventory, record damaged/returned books, pack orders and process warehouse fulfillment.
Customer	Browse books, search and filter books, manage cart, place customer orders, view order history and track deliveries.
Sales Manager	Monitor customer/wholesale sales, view orders, and approve or reject bookstore/wholesale orders.
Bookstore	Submit wholesale orders, view order status and track wholesale deliveries.
Delivery Staff	Handle assigned deliveries, update delivery status and record delivery completion.

EPIC 4 – Administration, Finance & Royalty Management

Table 6.4: Administration, Finance & Royalty Management

Swimlane	**Main responsibilities in the process**
Author	Receives royalty statement and payment; verifies figures against own records; confirms receipt or raises a dispute
Executive / Managing Director	Approve or reject high-value royalty payments, review financial reports, and monitor overall financial performance.

Administrator	Maintain author royalty agreement files, provide agreement details to Finance, review royalty calculations, and record and file completed payment records.
Finance Officer	Request and reconcile sales records, verify royalty agreements, calculate royalty amounts, prepare royalty statements, check fund availability, process royalty payments, and prepare financial reports.
Sales and Distribution Staff	Compile completed sales records for the royalty period from order books and spreadsheets and forward the sales summary to Finance.

7. Extract the process activities

EPIC 1 – Author & Manuscript Management

Table 7.1: Author & Manuscript Management

No.	**Activity**	**Responsible swimlane**
1	Prepare Manuscript	Author
2	Enter manuscript information	Author
3	Submit manuscript	Author
4	Provide Supporting Documents	Author
5	Receive Submitted Manuscript	Editorial Team
6	Record Manuscript Submission	Editorial Team
7	Review Manuscript	Editorial Team

8	Request Revision *(if required)*	Editorial Team
9	Receive Revision Request	Author
10	Revise Manuscript	Author
11	Resubmit Revised Manuscript	Author
12	Receive Revised Manuscript	Editorial Team
13	Re-review Manuscript	Editorial Team
14	Accept or Reject Manuscript	Editorial Team
15	Transfer accepted manuscript information to Epic 2	Editorial Team

EPIC 2 – Editorial Workflow & Book Production

Table 7.2: Editorial Workflow & Book Production

No.	**Activity**	**Responsible swimlane**
1	Receive accepted manuscript	Chief Editor
2	Assign editor	Chief Editor
3	Review manuscript	Editor
4	Send revision request *(if required)*	Editor
5	Receive revised manuscript	Chief Editor
6	Re-review manuscript	Editor
7	Approve editorial decision	Chief Editor

8	Create cover design	Designer
9	Prepare page layout	Designer
10	Revise design/layout *(if required)*	Designer
11	Perform quality check	Production Manager
12	Review book design	Author
13	Grant production approval	Production Manager
14	Mark book ready for printing	Production Manager
15	Transfer print-ready book to Printing Department	Production Manager

EPIC 3 – Publishing Operations (Printing, Inventory, Sales & Distribution)

Table 7.3: Publishing Operations

No.	**Activity**	**Responsible swimlane**
1	Create print job	Production Staff
2	Schedule printing	Production Staff
3	Print books	Production Staff
4	Check print quality	Production Staff
5	Complete print job	Production Staff
6	Record printed books	Production Staff
7	Receive printed books	Warehouse Staff
8	Update stock records	Warehouse Staff

9	Browse and search books	Customer
10	Add books to cart	Customer
11	Update cart	Customer
12	Checkout	Customer
13	Validate available stock	Warehouse Staff
14	Confirm customer order	Warehouse Staff
14	Create customer order	Customer
15	Pack customer order	Warehouse Staff
16	Dispatch order	Warehouse Staff
17	Assign delivery	Delivery Staff
18	Update delivery status	Delivery Staff
19	Record delivery completion	Delivery Staff
20	Submit wholesale order	Bookstore
21	Approve/reject wholesale order	Sales Manager
22	Reserve wholesale stock	Warehouse Staff
23	Pack and dispatch wholesale order	Warehouse Staff
24	Deliver wholesale order	Delivery Staff

EPIC 4 – Administration, Finance & Royalty Management

Table 7.4: Administration, Finance & Royalty Management

No.	Activity	Responsible swimlane
1	Initiate period-end royalty processing	Finance Officer
2	Request sales records for the royalty period	Finance Officer
3	Compile sales figures from order books and spreadsheets	Sales & Distribution Department
4	Forward sales summary to Finance	Sales & Distribution Department
5	Reconcile sales figures against dispatch and stock records	Finance Officer
6	Retrieve royalty agreement from the contract file	Administration Officer
7	Verify agreement validity and royalty percentage	Finance Officer
8	Calculate royalty amount manually	Finance Officer
9	Prepare royalty statement	Finance Officer
10	Review royalty calculation	Administration Officer
11	Forward high value payment for executive approval	Finance Officer
12	Approve or reject royalty payment	Managing Director
13	Check available funds against the payment amount	Finance Officer
14	Issue payment to author	Finance Officer
15	Send royalty statement and payment advice to author	Finance Officer
16	Verify statement and confirm receipt or raise dispute	Author

17	Record payment in ledger and file documents	Administration Officer
18	Compile period financial report	Finance Officer
19	Review financial report	Managing Director

8. Identify decision points

EPIC 1 – Author & Manuscript Management

Table 8.1: Author & Manuscript Management

Decision question	**Possible outcomes**
Has the manuscript been submitted with the required information and supporting documents?	Complete / Incomplete
Has a revision request been received for the manuscript?	Revision Requested / No Revision Requested
Has the author submitted the revised manuscript?	Revised Submitted / Revision Pending
Has the editorial team reached a final manuscript decision?	Accepted / Rejected / Revision Required

EPIC 2 – Editorial Workflow & Book Production

Table 8.2: Editorial Workflow & Book Production

Decision question	**Possible outcomes**
Is the manuscript acceptable after editorial review?	Accepted / Rejected / Revision Requested

Did the book pass the production quality check?	Passed / Corrections Required
Has the author approved the completed book design?	Approved / Changes Requested
Has the Production Manager granted final production approval?	Approved (Ready for Printing) / Corrections Required

EPIC 3 – Publishing Operations (Printing, Inventory, Sales & Distribution)

Table 8.3: Publishing Operations

Decision question	**Possible outcomes**
Is the book approved for printing?	Approved / Not Approved
Did the printed books pass quality checking?	Passed / Failed
Can the customer cancel the order?	Can cancel / Cannot cancel
Is the required book available in stock?	Available / Unavailable
Is the wholesale order approved?	Approved / Not Approved
Is the delivery status transition valid?	Valid / Invalid
Has the returned book passed inspection?	Pass / Fail
Was the delivery completed correctly?	Correct/ Incorrect

EPIC 4 – Administration, Finance & Royalty Management

Table 8.4: Administration, Finance & Royalty Management

Decision question	**Possible outcomes**
Are the sales records complete and reconciled?	Complete / Incomplete
Is a valid royalty agreement available on file?	Valid / Missing or Expired
Is the royalty calculation correct on review?	Correct / Error Found
Does the royalty amount exceed the approval threshold?	Above Threshold / Within Threshold
Has management approved the royalty payment?	Approved / Rejected
Are sufficient funds available for payment?	Sufficient / Insufficient
Has the author accepted the royalty statement?	Accepted / Disputed

9. Identify handovers and waiting points

EPIC 1 – Author & Manuscript Management

Table 9.1: Author & Manuscript Management

From	To	Information or Document transferred	Possible issue
Author	Editorial Team	Manuscript, manuscript information, and supporting documents	Missing information or incomplete document submission may delay manuscript processing.
Editorial Team	Author	Revision request and review comments	Delay in communicating or responding to the revision request may extend the manuscript processing time.
Author	Editorial Team	Revised manuscript and revision information	Incorrect or incomplete revised manuscript may require another revision cycle.
Editorial Team	Author	Editorial status and final manuscript outcome	Delayed communication may reduce visibility of manuscript progress

EPIC 2 – Editorial Workflow & Book Production

Table 9.2: Editorial Workflow & Book Production

From	To	Information or document transferred	Possible issue
Chief Editor	Editor	Assigned manuscript and editorial instructions	Delay in assigning an editor or incomplete assignment details

Editor	Chief Editor	Editorial review decision and review comments	Incomplete review comments or delayed review submission
Chief Editor	Designer	Approved manuscript for book production	Delay in commencing design activities
Designer	Author	Book cover design and page layout for review	Delay in author feedback or multiple design revisions
Author	Production Manager	Approved book design or requested design changes	Rework required if design changes are requested
Production Manager	Epic 3 – Publishing Operations	Print-ready book package (Book ID, ISBN, approved cover design, final layout, production approval)	Delay in handover or incomplete production package

EPIC 3 – Publishing Operations (Printing, Inventory, Sales & Distribution)

Table 9.3: EPIC 4 - Publishing Operations

From	**To**	**Information or document transferred**	**Possible issue**
Editorial Department	Production Staff	Printing approval and book details	Approval may be delayed or incomplete
Production Staff	Warehouse Staff	Printed books and print details	Books may be damaged or quantities may be incorrect
Customer	Sales Manager	Customer order details	Missing or incorrect order information

Bookstore	Sales Manager	Wholesale order	Order may require clarification or approval
Sales Manager	Warehouse Staff	Approved order	Delay in sending the order to the warehouse
Warehouse Staff	Delivery Staff	Packed order and delivery details	Incorrect address or incomplete delivery information
Delivery Staff	Customer	Books and delivery confirmation	Incorrect or damaged delivery

EPIC 4 – Administration, Finance & Royalty Management

Table 9.4: Administration, Finance & Royalty Management

From	**To**	**Information or document transferred**	**Possible issue**
Finance Officer	Sales and Distribution Staff	Request for period sales records	Request may be delayed or overlooked as there is no formal tracking
Sales and Distribution Staff	Finance Officer	Sales summary for the royalty period	Sales figures may be incomplete or inconsistent with dispatch records
Administrator	Finance Officer	Royalty agreement details and royalty rate	Agreement may be misfiled, expired, or an

			outdated version may be supplied
Finance Officer	Administrator	Royalty calculation and draft royalty statement	Review may be delayed and calculation errors detected late
Finance Officer	Executive / Managing Director	High-value royalty payment approval request	Approval may be delayed while awaiting the availability of management
Executive / Managing Director	Finance Officer	Approved or rejected payment decision	Rejection reasons may be unclear or approval documents may be misplaced
Finance Officer	Author	Royalty statement and payment advice	Statement may be delayed in transit and the author cannot verify the underlying sales figures
Author	Finance Officer	Payment confirmation or royalty dispute	Disputes may be raised late and communicated informally without a proper record
Finance Officer	Administrator	Completed payment record for filing	Duplicate or missing entries may occur in the manual financial records

10. Draw the As-Is UML Activity Diagram

EPIC 1 – Author & Manuscript Management

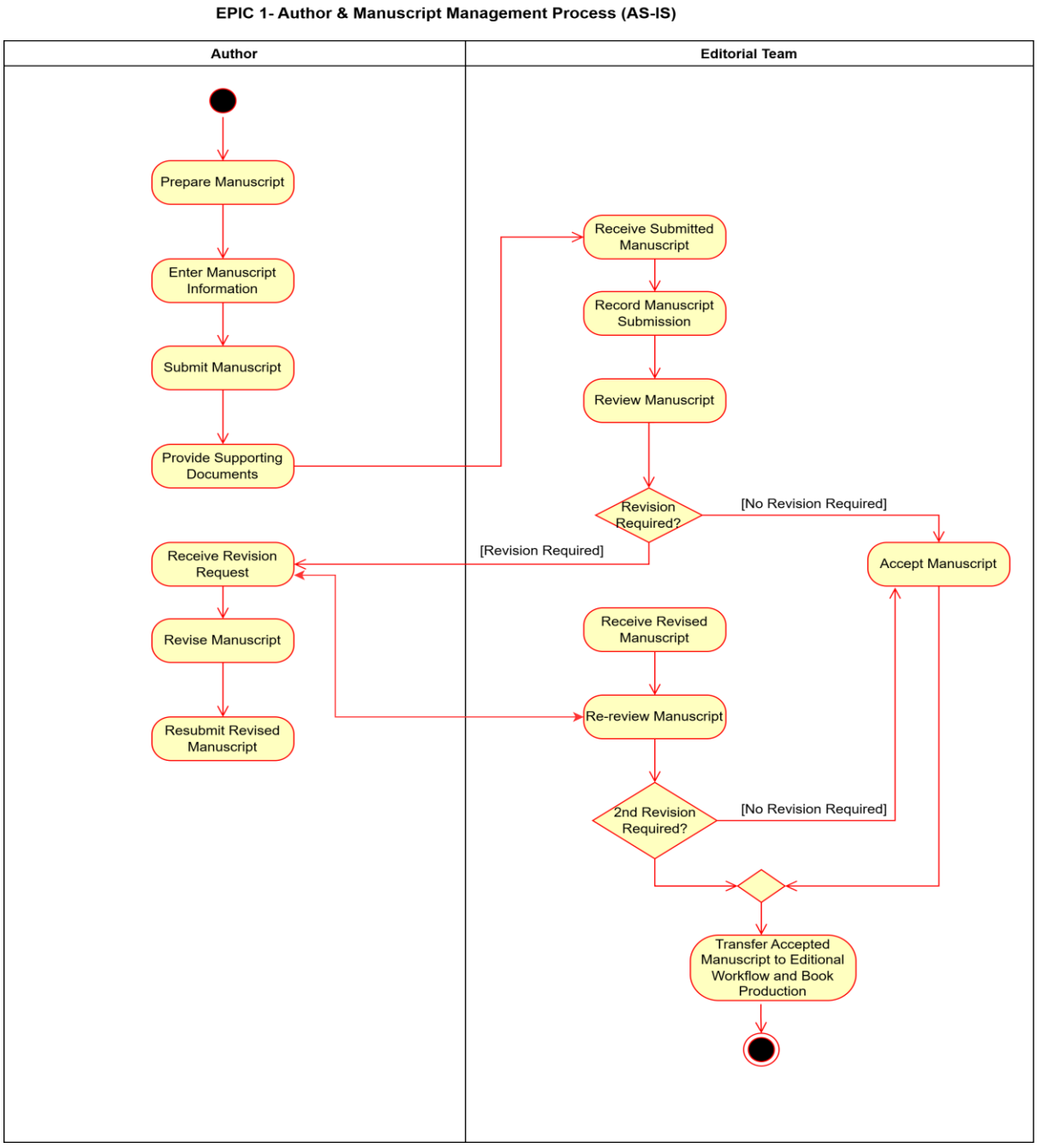


Figure 10.1: Author & Manuscript Management As-Is UML Activity Diagram

EPIC 2 – Editorial Workflow & Book Production

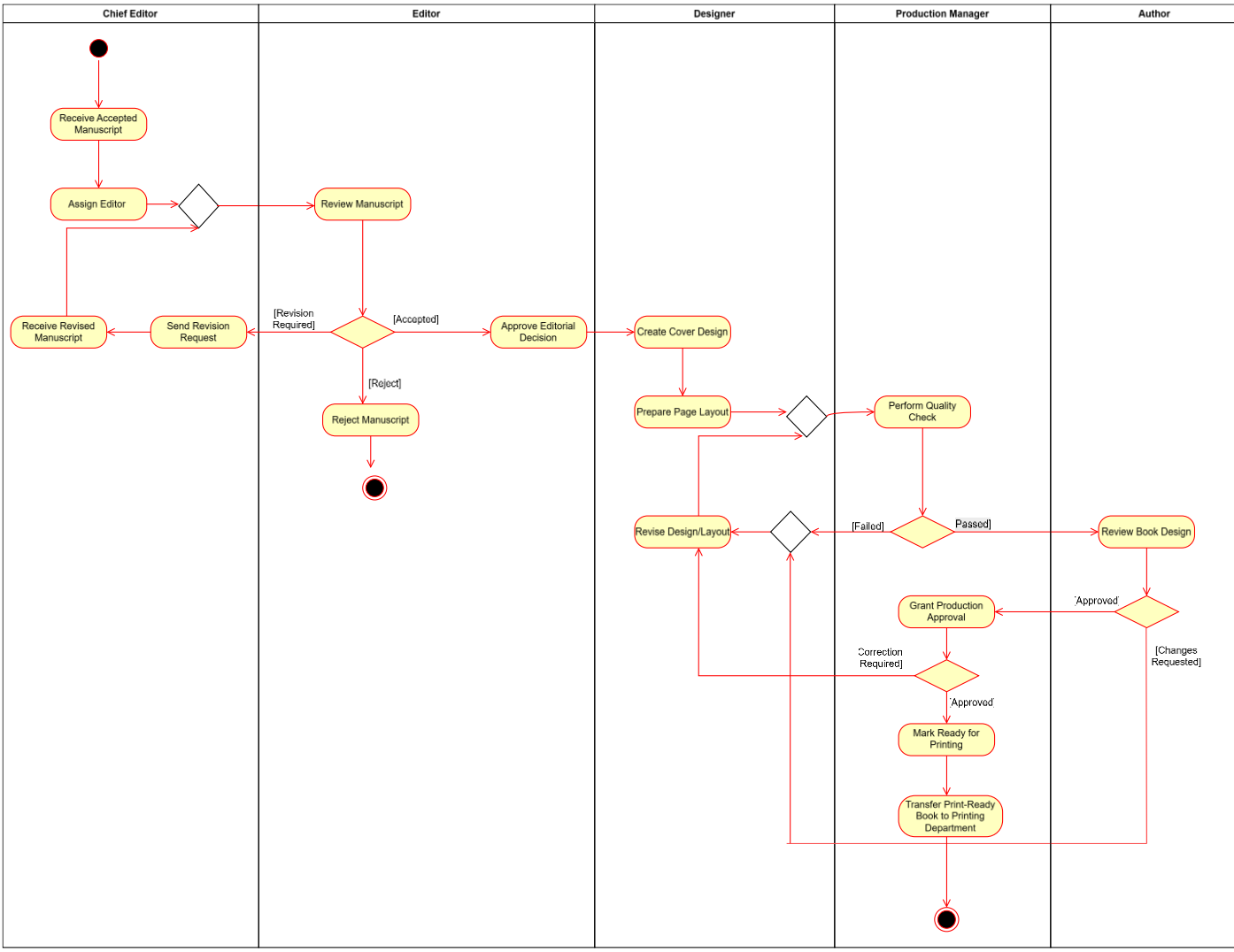


Figure 10.2 : Editorial Workflow & Book Production As-Is UML Activity Diagram

EPIC 3 – Publishing Operations (Printing, Inventory, Sales & Distribution)

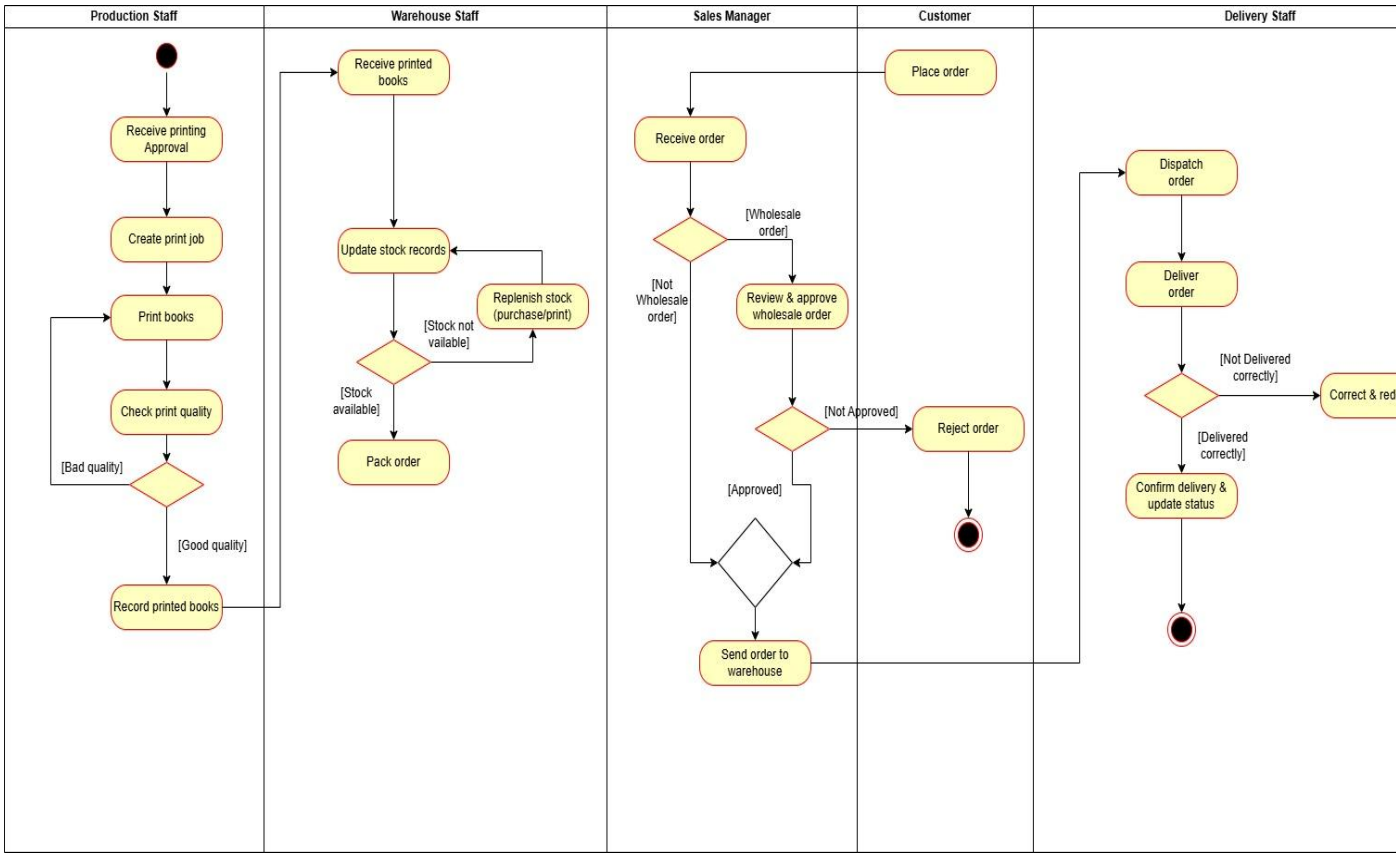


Figure 10.3: Publishing Operations As-Is UML Activity Diagram

EPIC 4 – Administration, Finance & Royalty Management

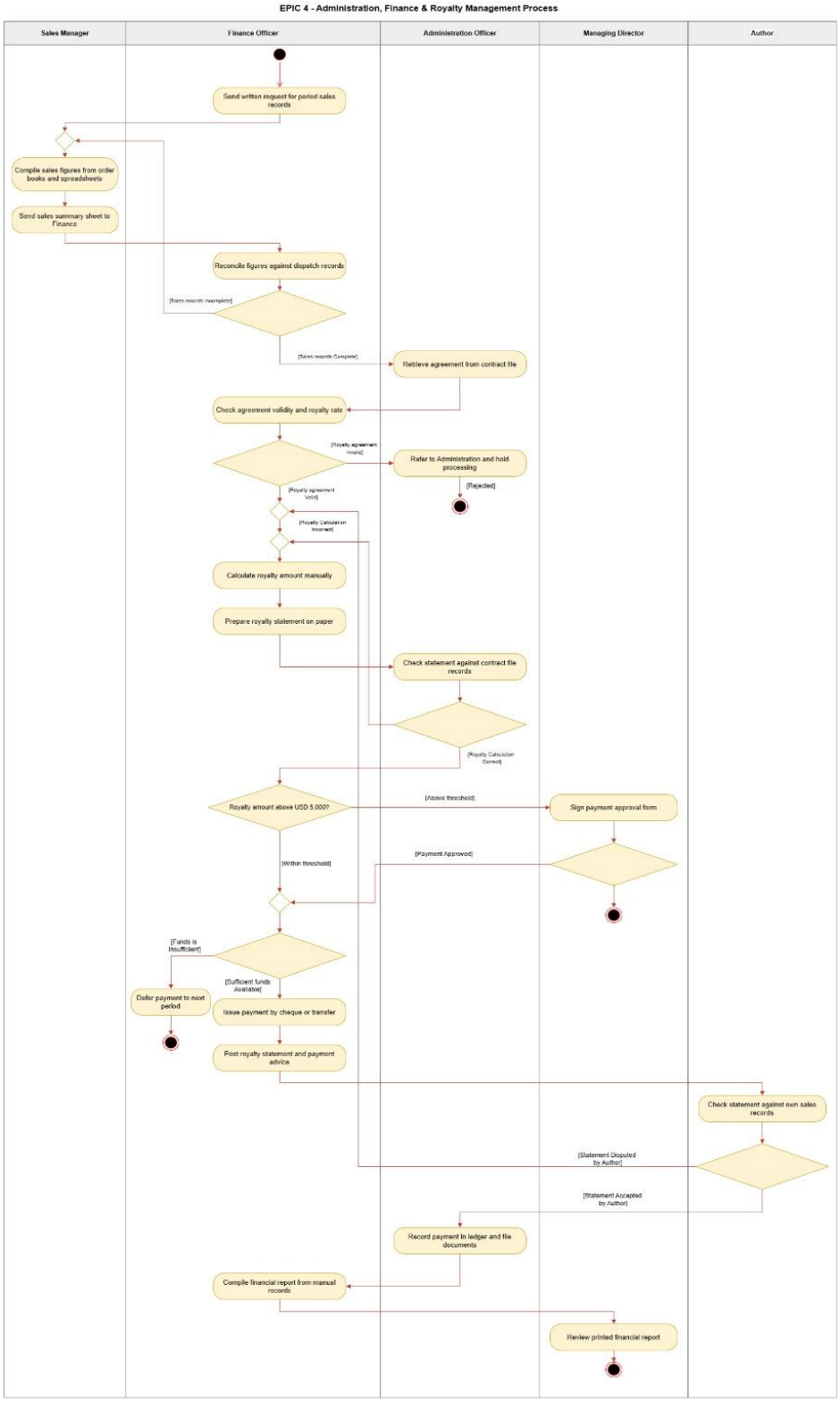


Figure 10.4: Author & Manuscript Management As-Is UML Activity Diagram

11. Validate the diagram

EPIC 1 – Author & Manuscript Management (Validate by IT24101035– Sewwandi R.P.D.D.C)

Table 11.1: Author & Manuscript Management

Validation question	**Yes**	**No**	**Comments**
Is the start of the process clear?	✓		Starts with Prepare Manuscript using an initial node.
Are all major actors represented as swimlanes?	✓		The Author and Editorial Team are represented as the main actors involved in the current manuscript management process.
Is every activity placed in the correct swimlane?	✓		Author activities are placed in the Author swimlane, while manuscript receiving, recording, reviewing, revision handling, and acceptance activities are placed in the Editorial Team swimlane.
Are activity labels clear and action-oriented?	✓		Activities use clear verb–object phrases such as Prepare Manuscript, Submit Manuscript, Review Manuscript, Revise Manuscript, and Accept Manuscript.
Are decision outcomes shown using guard conditions?	✓		Guard conditions such as [Revision Required] and [No Revision Required] are used for the manuscript review decisions.
Are rejection paths represented?		×	A separate rejection path is not represented because the Epic 1 process does not define a manuscript rejection

			activity. Manuscripts requiring changes follow the revision and re-review path.
Is the stock-availability decision represented?		×	Not applicable to the Author & Manuscript Management process.
Is the USD 5,000 approval threshold represented?		×	Not applicable to the Author & Manuscript Management process.
Is the insufficient-budget path represented?		×	Not applicable to the Author & Manuscript Management process.
Is the incorrect-delivery loop represented?		×	Not applicable to the Author & Manuscript Management process.
Are the end points clearly shown?	✓		An activity final node is used after the accepted manuscript is transferred to Epic 2 – Editorial Workflow & Book Production.
Does the diagram represent the scenario accurately?	✓		The diagram represents the current author and manuscript management process, including manuscript submission, editorial review, revision and re-review cycles, acceptance, and handover to Epic 2.

EPIC 2 – Editorial Workflow & Book Production (Validate by IT24100696 – Dissanayake L.N)

Table 11.2 : Editorial Workflow & Book Production

Validation question	**Yes**	**No**	**Comments**
Is the start of the process clear?	✓		Starts with Receive Accepted Manuscript using an initial node.
Are all major actors represented as swimlanes?	✓		Chief Editor, Editor, Designer, Production Manager, and Author are represented.
Is every activity placed in the correct swimlane?	✓		Activities are assigned to the appropriate stakeholder responsible for performing them.
Are activity labels clear and action-oriented?	✓		Activities use verb–object phrases (e.g., *Review Manuscript*, *Design Book Cover*).
Are decision outcomes shown using guard conditions?	✓		Guard conditions such as **[**Accepted]**,** [Rejected]**,** [Revision Required]**,** [Passed]**,** [Failed]**,** [Approved], and [Changes Requested] are used.
Are rejection paths represented?	✓		Editorial rejection terminates the process and author-requested changes create a rework loop.
Is the stock-availability decision represented?		×	Not applicable to the editorial publishing process.
Is the USD 5,000 approval threshold represented?		×	Not applicable to this business process

Is the insufficient-budget path represented?		×	Not applicable to the editorial publishing process.
Is the incorrect-delivery loop represented?		×	Not applicable to the editorial workflow.
Are the end points clearly shown?	✓		Separate final nodes are used for manuscript rejection and successful handover to Publish Operations (Epic 3).
Does the diagram represent the scenario accurately?	✓		The diagram accurately models the current manual editorial and production workflow described in Epic 2.

EPIC 3 – Publishing Operations (Validate by IT24101357– Bandara N.H.M.W.W.D.S)

Table 11.3: Publishing Operations

Validation question	**Yes**	**No**	**Comments**
Is the start of the process clear?	✓		Process starts when printing approval is received.
Are all major actors represented as swimlanes?	✓		Production, Warehouse, Customer, Sales Manager, Bookstore and Delivery are included.
Is every activity placed in the correct swimlane?	✓		Activities are assigned according to responsibilities.
Are activity labels clear and action-oriented?	✓		Activities use verb–object phrases.

Are decision outcomes shown using guard conditions?	✓		Guard conditions such as [Approved], [Rejected], [Available], [Unavailable], etc. are shown.
Are rejection paths represented?	✓		Rejected wholesale/purchase requests are represented.
Is the stock-availability decision represented?	✓		Available and unavailable paths are included.
Is the USD 5,000 approval threshold represented?		×	Not applicable to the process
Is the insufficient-budget path represented?		×	Not applicable to the process
Is the incorrect-delivery loop represented?	✓		Incorrect deliveries are corrected and redelivered.
Are the end points clearly shown?	✓		Successful delivery and rejected/terminated paths have final nodes.
Does the diagram represent the scenario accurately?	✓		The diagram follows the defined Epic 3 process.

EPIC 4 – Administration, Finance & Royalty Management (Validate by IT24100863– Ananda U.V.D.M.A)

Table 11.4: Administration, Finance & Royalty Management

Validation question	**Yes**	**No**	**Comments**
Is the start of the process clear?	✓		
Are all major actors represented as swimlanes?	✓		
Is every activity placed in the correct swimlane?	✓		
Are activity labels clear and action-oriented?	✓		
Are decision outcomes shown using guard conditions?	✓		
Are rejection paths represented?	✓		
Is the stock-availability decision represented?		×	
Is the USD 5,000 approval threshold represented?	✓		
Is the insufficient-budget path represented?	✓		
Is the incorrect-delivery loop represented?		×	
Are the end points clearly shown?	✓		
Does the diagram represent the scenario accurately?	✓		

12. Analyze the current process

EPIC 1 – Author & Manuscript Management

Table 12.1: Author & Manuscript Management

No.	Problem identified	Evidence from the process	Business impact
1	Manual manuscript submission	Authors currently rely on manual methods such as paper documents, emails, and other fragmented methods to submit manuscripts and supporting files.	Increases administrative workload and the risk of missing or incomplete submission information.
2	Limited submission status visibility	Authors cannot easily track the current progress of their manuscript through the existing manual process.	Reduces transparency and increases uncertainty for authors regarding manuscript progress.
3	Manual revision handling	Revision requests and revised manuscript submissions require communication and document exchange between the publishing house and the author.	Increases processing time and may delay the completion of the manuscript submission cycle.
4	Scattered manuscript information	Manuscript details, supporting documents, submission information, and communication may be maintained across different manual channels.	Makes information retrieval and coordination more difficult and increases the risk of information being misplaced or inconsistent.

5	Difficulty managing manuscript versions	The revision cycle involves submitting revised manuscript versions through the existing manual communication and document submission methods, making it difficult to maintain and identify the latest version.	Makes it more difficult to identify the latest manuscript version and can lead to confusion during manuscript processing.

EPIC 2 – Editorial Workflow & Book Production

Table 12.2: Editorial Workflow & Book Production

No.	**Problem identified**	**Evidence from the process**	**Business impact**
1	Manual editor assignment	The Chief Editor manually assigns each accepted manuscript to an editor before the review begins.	Delays the start of the review process and increases administrative workload.
2	Lack of centralized revision tracking	Manuscripts may go through multiple revision rounds, making it difficult to track previous comments, versions, and outstanding changes.	Increases the risk of missed feedback, version confusion, and delays in editorial completion.
3	Manual approval coordination	Editorial, author, and production approvals occur at different	Creates waiting periods and makes approval status difficult to monitor.

		stages and require coordination between stakeholders.	
4	Multiple departmental handovers	The manuscript passes through the Chief Editor, Editor, Designer, Author, and Production Manager.	Creates coordination delays and increases the risk of process bottlenecks.
5	Limited process visibility	The workflow involves several departments and repeated handovers, making it difficult to track the status of a manuscript.	Reduces process transparency, making monitoring and progress reporting more difficult.

EPIC 3 – Publishing Operations (Printing, Inventory, Sales & Distribution)

Table 11.3: Publishing Operations

No.	**Problem identified**	**Evidence from the process**	**Business impact**
1	Manual stock management	Stock information is recorded and checked manually.	Stock discrepancies and stock-out situations may occur.
2	Slow approval process	Wholesale and high-value orders require manual approval.	Orders may be delayed.
3	Poor information sharing	Information moves between Production, Warehouse, Sales and Delivery.	Employees may not have up-to-date information.

4	Manual order tracking	Customer and bookstore orders are handled through separate records and communication.	Difficult to determine order status quickly.
5	Delivery errors	Incorrect deliveries require correction and redelivery.	Extra time and delivery costs are incurred, and customer satisfaction may decrease.

EPIC 4 – Administration, Finance & Royalty Management

Table 11.4: Author & Manuscript Management

No.	**Problem identified**	**Evidence from the process**	**Business impact**
1	Manual collection of sales data	The Finance Officer must request sales records from Sales and Distribution, which compiles them by hand from order books and spreadsheets.	The royalty cycle cannot begin on time and processing is delayed at the start of every period.
2	Manual royalty calculation	Royalty amounts are calculated by hand using sales figures and agreement rates, with a rework loop when errors are found at review.	Calculation errors cause rework, delay payments, and may result in incorrect amounts being paid to authors.
3	Delayed approval of payments	High-value royalty payments require manual approval from the Executive/Managing Director.	Payments are delayed while awaiting approval and the status of pending approvals is not visible.

4	Scattered financial and agreement records	Royalty agreements are held by the Administrator, sales figures by Sales, and payment records in manual financial records.	Information is difficult to retrieve, outdated agreement versions may be used, and records may become inconsistent.
5	Limited royalty visibility for authors	Authors receive royalty information only when the statement is issued and cannot verify figures independently.	Disputes are raised late, requiring recalculation and reducing author satisfaction and trust.
6	No real-time financial reporting	Financial and revenue reports are compiled manually after the royalty period closes.	Management decisions are based on outdated financial information.

13. Identify measurable baseline indicators

EPIC 1 – Author & Manuscript Management

Table 13.1: Author & Manuscript Management

KPI	**How it could be measured**
Average manuscript submission processing time	Average number of days from the start of manuscript submission until the manuscript is successfully submitted with all required information and documents.
Average revision response time	Average number of days from a revision request being received by the author until the revised manuscript is resubmitted.
Percentage of manuscripts requiring revision	(Number of manuscripts for which revision is requested ÷ Total manuscripts submitted) × 100.

Average number of manuscript revisions	Total number of revised manuscript submissions ÷ Total manuscripts that required revision.
Average manuscript acceptance cycle time	Average number of days from initial manuscript submission until the manuscript reaches the **Accepted** status.

EPIC 2 – Editorial Workflow & Book Production

Table 13.2: Editorial Workflow & Book Production

KPI	**How it could be measured**
Average manuscript review time	Average number of days from editor assignment to the editorial decision (accepted, rejected, or revision required
Average revision cycle time	Average number of days from requesting manuscript revisions until the revised manuscript is received.
Percentage of manuscripts requiring revision	(Number of manuscripts returned for revision ÷ Total manuscripts reviewed) × 100.
Average author design approval time	Average number of days from completion of the book design until the author approves the design.
Average production preparation time	Average number of days from manuscript approval to marking the book as Ready for Printing.

EPIC 3 – Publishing Operations (Printing, Inventory, Sales & Distribution)

Table 11.3: Publishing Operations

KPI	**How it could be measured**
Average print job completion time	Measure the time from creating a print job until printing is completed.
Percentage of stock-record discrepancies	Compare physical stock with the recorded stock and calculate the percentage of differences.
Average order processing time	Measure the time from receiving an order until it is packed and ready for dispatch.
Percentage of deliveries requiring correction	Number of incorrect deliveries ÷ total deliveries × 100.
Average time taken to determine order status	Measure how long staff take to find and provide the current status of an order.

EPIC 4 – Administration, Finance & Royalty Management

Table 13.4 – Administration, Finance & Royalty Management

KPI	**How it could be measured**
Average royalty processing cycle time	Average number of days from the end of the royalty period until the royalty payment is completed.
Average payment approval time	Average number of days from forwarding a high-value payment for approval until the approval decision is received.

Percentage of royalty calculations requiring correction	Number of calculations returned for correction ÷ Total calculations performed × 100.
Percentage of royalty statements disputed by authors	Number of disputed royalty statements ÷ Total royalty statements issued × 100.
Percentage of sales summaries returned as incomplete	Number of sales summaries returned for correction ÷ Total sales summaries received × 100.
Average financial report preparation time	Average number of days from the end of the financial period until the report is completed and reviewed.
Number of manual data-entry points per royalty cycle	Count of activities in which financial figures are manually copied or re-entered from one record into another.

14. Prepare a brief process assessment

EPIC 1 – Author & Manuscript Management

The current author and manuscript management process begins when an author prepares a manuscript and provides the required manuscript information and supporting documents for submission. The manuscript is submitted to the publishing house and received by the Editorial Team for further processing. The author may track the progress of the manuscript and respond when a revision is requested by preparing and resubmitting a revised manuscript version. The manuscript may go through repeated revision cycles before an editorial outcome is reached. Once the manuscript is accepted, it proceeds through the subsequent editorial and book production processes.

The major delays in the current process are associated with manual manuscript submission, limited visibility of submission progress, manual revision handling, scattered manuscript information, and difficulty managing different manuscript versions. These issues increase

the time required to complete the manuscript submission lifecycle and create additional administrative effort for both authors and the publishing house. Authors are particularly affected because they have limited visibility into the progress of their submitted manuscripts and may need to rely on manual communication when revisions are required.

Activities such as manual document exchange, repeated communication regarding revisions, and searching for manuscript information add administrative effort without directly improving the value of the manuscript. Priority should therefore be given to improving manuscript submission coordination, increasing submission status visibility, centralizing manuscript information and supporting files, and improving revision and version management. Before designing the To-Be process, evidence should be collected on manuscript submission processing times, revision turnaround times, number of revision cycles, manuscript acceptance cycle times, and delays in accessing or managing manuscript information. These measurements will provide a reliable baseline for evaluating future improvements to the Author & Manuscript Management process.

EPIC 2 – Editorial Workflow & Book Production

The current editorial workflow begins when the Chief Editor receives an accepted manuscript from Epic 1 and assigns it to an editor for review. The editor reviews the manuscript and decides whether to accept it, reject it, or request revisions. If revisions are required, the revised manuscript is received and reviewed again until it satisfies the editorial requirements. Once approved, the manuscript proceeds to book cover design and page layout preparation. The Production Manager performs a quality check before the author reviews the completed book design. If the author requests changes, the design is revised and submitted for review again. After the author approves the design, the Production Manager grants production approval, marks the book as Ready for Printing, and transfers the approved files to Publishing Operations.

The major delays occur during editor assignment, repeated manuscript revision cycles, multiple approval stages, and handovers between departments. These delays increase the overall time required to prepare a book for printing. The Chief Editor, Editor, Designer,

Author, and Production Manager are the stakeholders most affected because each depends on the completion of activities performed by others.

Priority should therefore be given to improving coordination between departments, reducing unnecessary waiting time, and increasing visibility of manuscript progress. Before designing the To-Be process, evidence should be collected on manuscript review times, revision frequency, approval turnaround times, handover delays, and overall production preparation time. These measurements will provide a reliable baseline for evaluating future process improvements.

EPIC 3 – Publishing Operations (Printing, Inventory, Sales & Distribution)

The whole of the publishing operations process has several stakeholders engaged in the task of getting books from the printing press to the inventory and then to the sales, order fulfillment and delivery. Printed books are handed over to Warehouse Staff for stock management after being created and completed by Production Staff. Customers may order books and place orders and wholesale orders will require Sales Manager approval for fulfillment. Stocking is done by Warehouse Staff, packing and despatching are done by Warehouse Staff and Delivery Staff completes the delivery process. Coordination and handovers between various stakeholders is therefore a crucial part of the process.

The primary delays are expected to be in the stock checking, wholesale order approval, high-value order approval and manual handover/delivery corrections. Warehouse Staff are especially impacted as it involves handling the stock, packing and fulfillment activities. When orders are pending to be stocked or approved, it impacts Customers and Bookstore/Wholesale Customers. When several sales orders must be reviewed manually, Sales Managers can suffer delays.

The things that they do in the process, like checking stocks again and again, shuffling data from one department to another, double entry, fixing delivery issues, etc., are not added value to the customer. They primarily assist the process and prolong the processing time.

So the key areas of improvement should be inventory visibility, order processing, approval handling, inter-department communication and order tracking for delivery. Prior to

defining the To-Be process, evidence should be gathered of actual processing times, approval times, discrepancies in stock, orders rejected, orders delivered incorrectly, number of times passed to other people, and frequency of rework. These measurements can be used as a dependable standard to test the enhanced process.

EPIC 4 – Administration, Finance & Royalty Management

The current administration system, finance and royalty management is completely based on manual records and exchange of physical documents. The Finance Officer asks the Sales and Distribution Department to prepare sales figures at the end of every royalty period, and they prepare it manually by adding up the sales of the order books and spreadsheets. The Finance Officer then takes the reconciliation of these numbers, requests the royalty agreement from the Administrator, checks the royalty rate, and manually works out the figure for the royalty, before producing a royalty statement. The Administrator checks the calculation and payments over the approved amount are sent to the Executive/Managing Director for approval. When approved and money is determined as available, the payment is made, the statement is sent to the author and the statement is noted in the financial records. Management review is then done manually in financial reports. Major delays are in the handover points not in the work itself. The waiting periods are caused by the compilation of sales records, retrieval of royalty agreement files, waiting for the calculation review, and waiting for management approval.

Authors are stakeholders who are most impacted because they do not receive their payments on time, do not have the ability to check the number of their royalties on their own and need to contact the publisher to find out what has happened to their payment. Sales data incompleteness and calculation errors also directly impact finance staff as they have to deal with the rework.

Re-entering sales data that is already inputted in other systems, retrieval of agreement files, follow up on pending approvals, and manual creation of reports provide little incremental value to the business and instead are administrative tasks.

There should then be a focus on enhancing the sales data collection process, the accuracy of royalty calculation and the speed of the approval of royalty payments. To create a

reliable base line for the To-Be process, evidence should be gathered on the following: Royalty cycle times, financial report preparation times, rates of calculation errors, rates of approval turnaround and dispute rates.

Appendix A

Meeting Notes

Meeting No.	**Date**	**Participants**	**Topics Discussed**	**Decisions / Outcomes**
1	24 July 2026	All group members	Project selection and client business problems	EPMS project and publishing house selected.
2	26 July 2026	All group members	Discussed the selected Book Publishing House, its business activities, and current operational processes.	Book Publishing House was selected as the client organization.
3	28 July 2026	All group members	Identified the client's main business problems and areas where digital transformation could provide value.	Agreed that the client's manual and disconnected processes required an integrated digital solution.
4	1 August 2026	All group members + Client Representative	Discussed the current business processes, operational problems, system requirements, and proposed project concept with the client.	Client requirements and business needs were clarified, and the proposed EPMS concept was discussed with the client.

5	3 August 2026	All group members + Client Representative	Presented the proposed Enterprise Publishing Management System and discussed its intended scope and major business functions.	Client approval/agreement was obtained to proceed with the proposed system concept.
6	5 August 2026	All group members	Reviewed the project scope and divided the system into four major epics.	Four epics were identified, and responsibilities were allocated among the four team members.
7	14 August 2026	All group members	Reviewed the Sprint 0 requirements, business processes, stakeholders, and initial project documentation.	Initial Sprint 0 documentation and project scope were agreed upon for further development.